

INSTITUTIONAL Portfolio Optimizer

[Link of Github](#)

STRATEGIC Portfolio GENERATION

Advanced Mathematical Modeling for Institutional Portfolios

EXECUTIVE SUMMARY & PORTFOLIO SCORECARD

STRATEGIC INVESTMENT PHILOSOPHY & METRIC REALIZATION

- Risk is allocated first; capital follows. The primary objective is maximizing risk-adjusted returns strictly within predefined boundaries.
- By anchoring the portfolio to a 14.00% volatility budget, the quantitative engine generated an expected return of 13.37%, achieving +6.54% Alpha.
- A strategic trade-off was executed regarding Market Beta: achieving the >0.90 target would have breached the volatility ceiling, so lower Beta was accepted.
- This prioritization of defensive compounding resulted in superior downside capital preservation, locking the maximum historical drawdown to -14.31%.

13.37%

RETURN FORECAST

13.30%

PORTFOLIO VOLATILITY

-14.31%

MAX DRAWDOWN

+6.54%

EXPECTED ALPHA

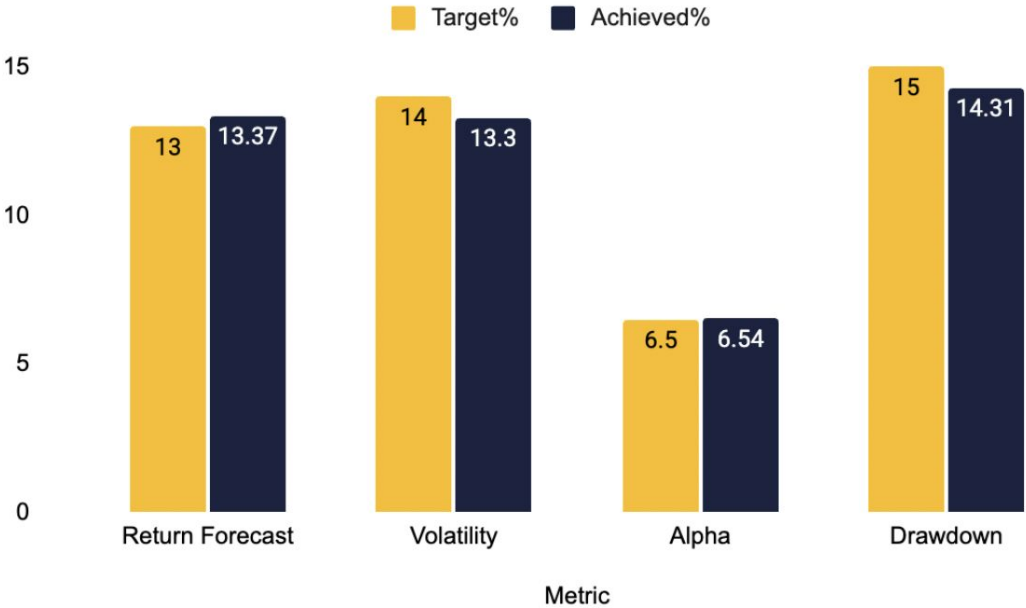
0.830

PORTFOLIO BETA

TARGET VS ACHIEVED SCORECARD

Metric	Target	Achieved	Status
Return Forecast	12-14%	13.37%	Target Met
Volatility	< 14.00%	13.30%	Target Met
Max Drawdown	> -15.00%	-14.31%	Target Met
Market Beta	0.85 - 0.95	0.830	Trade-Off

Target and Achieved Metrics



METHODOLOGY: MODELS, ASSUMPTIONS & LIMITATIONS

MATHEMATICAL ARCHITECTURE & MODEL ENGINE

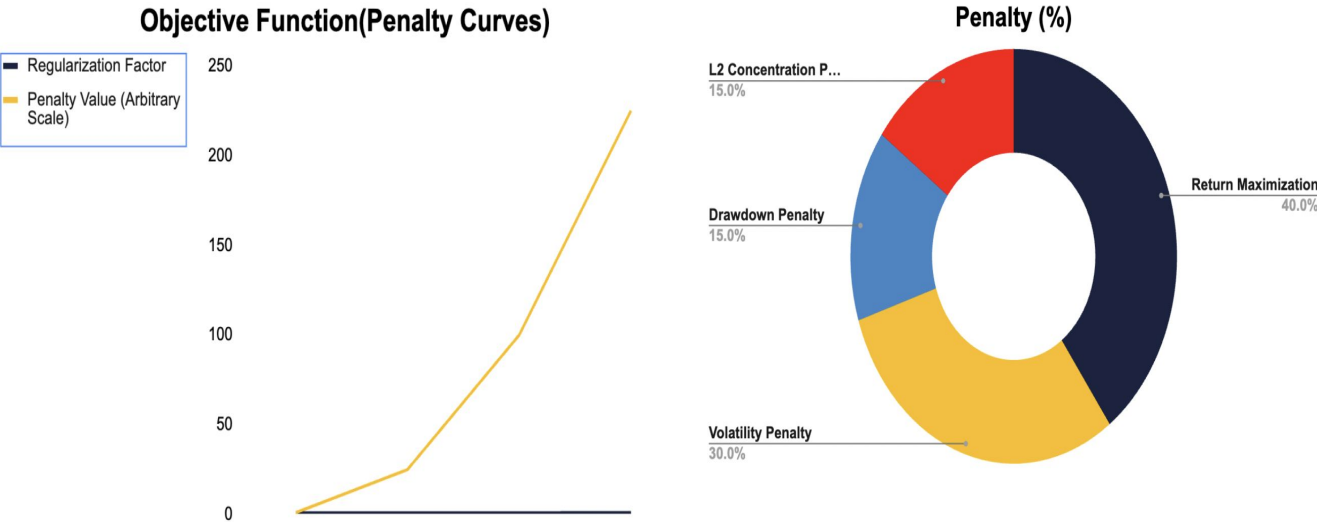
- Primary Solver (SLSQP): Evaluates non-linear risk/return objectives subject to multi-layered portfolio limits.
- Fallback Engine (Trust-Constr): Automatically deployed to handle complex constraint geometries if gradient-methods fail.
- L2 Regularization: Integrates mathematically smoothed quadratic gravity wells to prevent error maximization.

KNOWN LIMITATIONS & BOUNDARY CONDITIONS

- Black Swan Vulnerability: Backward-looking covariance cannot anticipate unprecedented exogenous tail-risk events.
- Frictionless Execution: Optimization assumes zero liquidity slippage or transaction taxation during rebalancing.
- Target Rigidity: Strict 14% volatility caps suppress the engine from opportunistic scaling in heavy bull regimes.

METHODOLOGY & ASSUMPTIONS MAPPING

Framework Component	Core Assumption	Impact on Output
Pre-Filtration	ROE accurately proxies structural strength	Eliminates value traps early
Covariance Math	Historical stationarity persists near-term	Drives asset weighting
Value at Risk (VaR)	Market returns exhibit normal distribution	Quantifies tail exposure



BENCHMARK ALIGNMENT & DIVERGENCE ANALYSIS

QUALITATIVE CONVICTION VS QUANTITATIVE ALLOCATION

- Investment research generates ideas, but portfolio construction creates the final investable product by mathematically allocating capital.
- When comparing the engine's 16 output holdings to a 16-asset qualitative Advisory Universe(One of the top Advisories) the model achieved a 37.5% exact overlap (6 shared names).
- This alignment confirms the optimizer inherently recognizes and selects for fundamentally strong assets without human bias.
- The divergence demonstrates institutional value: the engine ruthlessly replaced correlated human picks with risk-efficient assets to drop aggregate Beta.

37.5%

ALIGNMENT RATIO

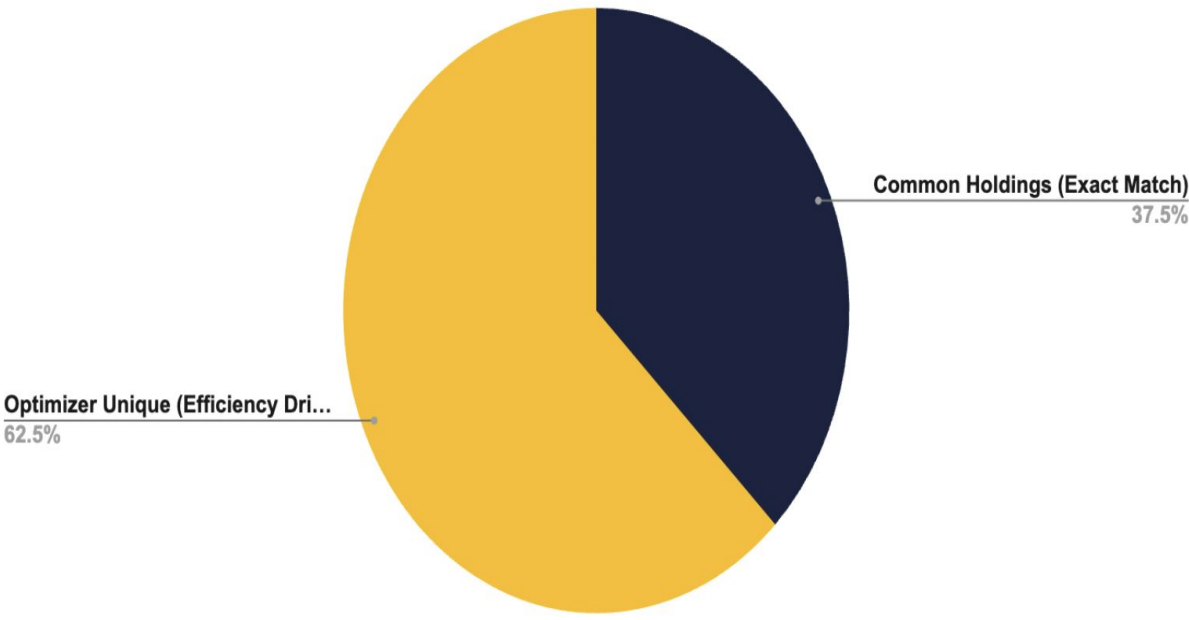
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COMMON HOLDINGS

DIVERGENCE RATIONALE MATRIX

Zone	Holdings	Interpretation / Rationale
Agreement	6 Stocks	Strong conviction — optimizer & analyst agree.
Optimizer Only	10 Stocks	Optimizer found superior covariance profiles.
Advisory Only	10 Stocks	Rejected due to high risk-adjusted capital cost.

Alignment Breakdown Percentage



PORTFOLIO CONSTRUCTION ARCHITECTURE

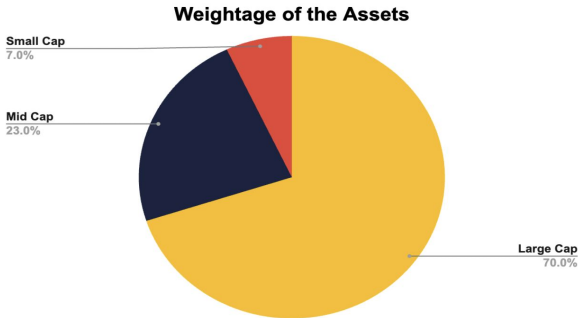
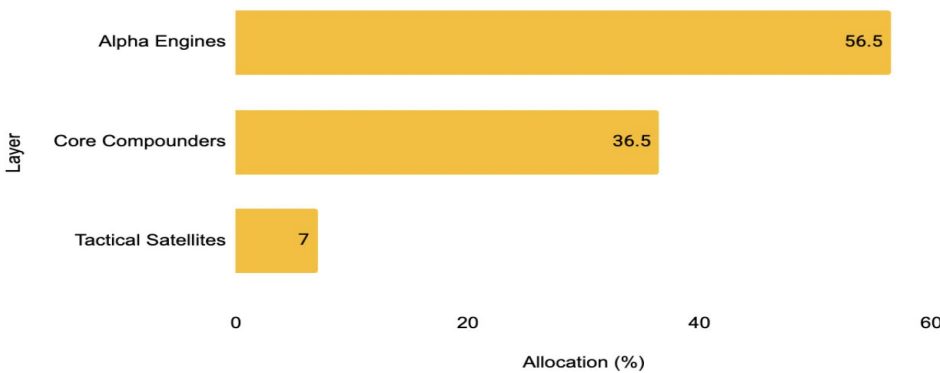
DIVERSIFICATION BY FUNCTION, NOT BY SECURITY COUNT

- Every security introduced into the portfolio serves a predefined mathematical role. The architecture is split into three distinct risk-weighted layers.
- Layer 1: Core Compounders (~36.5%). Low beta, high-quality fundamentals (e.g., Marico, HDFC Bank). Acts as the portfolio's volatility anchor.
- Layer 2: Alpha Engines (~56.5%). Primary return generators (e.g., Bharti Airtel, UltraTech). Sized heavily due to superior forward Sharpe ratios.
- Layer 3: Tactical Satellites (~7.0%). High beta cyclical exposures (e.g., JSW Steel) strictly capped to participate in upside without ruining drawdown metrics.

STRUCTURAL LAYER TARGETS

Portfolio Layer	Allocation	Target Beta	Fundamental Role
Core Compounders	~36.5%	0.60 - 0.80	Volatility Anchor / Stability
Alpha Engines	~56.5%	0.85 - 1.05	Primary Return Generation
Tactical Satellites	~7.0%	1.10 - 1.35	Controlled Upside Cyclicalty

Allocation Percentage of the Portfolio



QUALITY CONTROL PRECEDES OPTIMIZATION

- Optimization begins only after exhaustive quality control. The optimizer is strictly forbidden from allocating capital to structurally weak businesses.
- The system scores every security in the Nifty 500 across five dimensions: Operating Margin, Revenue Growth, Debt-to-Equity, ROE, and ROCE.
- A strict quality threshold is enforced: Any security with a Fundamental Score below 0.45 is automatically eliminated from the investable universe.
- This process acts as a downside floor, ensuring that even if correlation math fails during a crash, the underlying assets possess intrinsic balance sheet strength.

0.694

PORTFOLIO AVG Fund Score

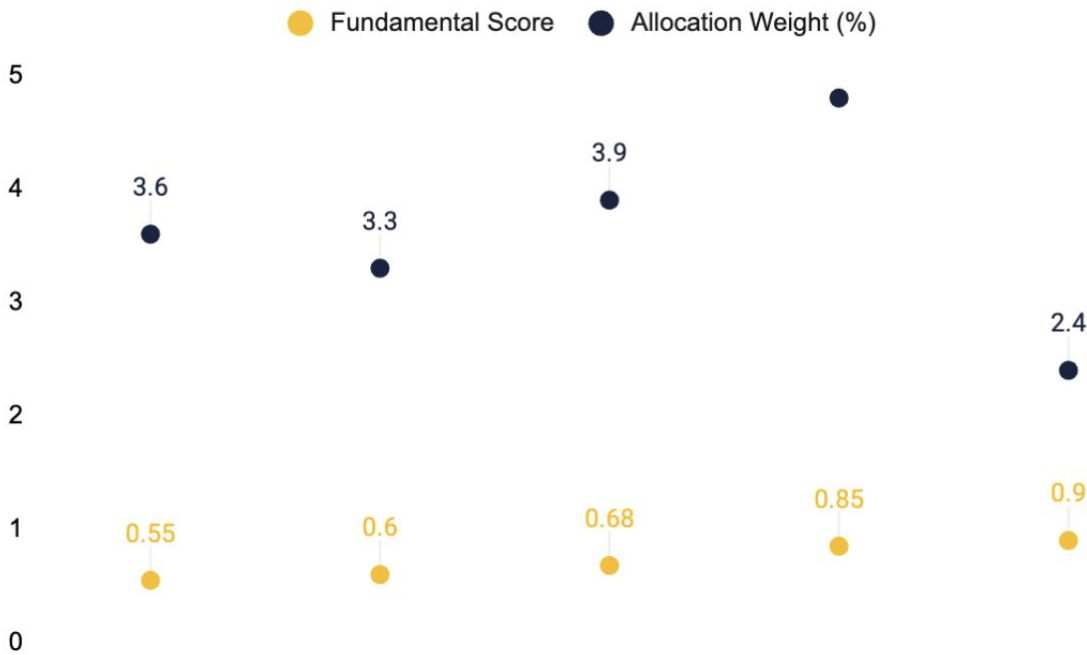
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SURVIVING ASSETS

QUALITY FILTRATION METRICS

Fundamental Metric	Financial Rationale	Target Trajectory
Operating Margin	Profitability efficiency	Top quartile sector relative
Debt-to-Equity	Balance sheet & leverage	Below industry average
Return on Equity	Shareholder value creation	Consistent > 15% generation

Fundamental Score and Allocation Weight (%)



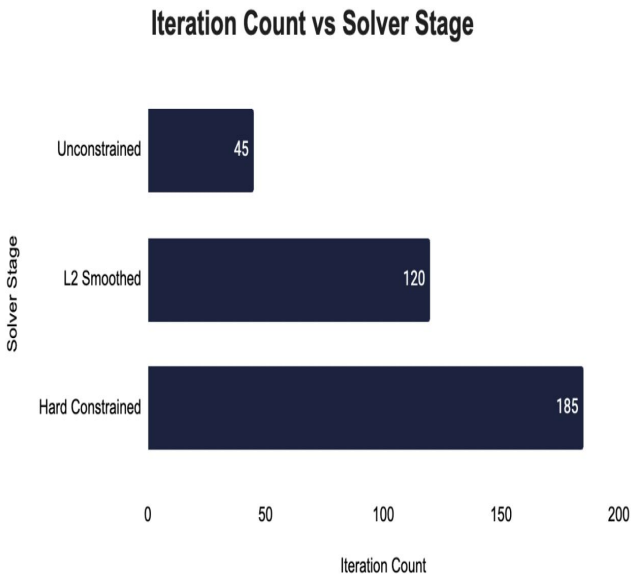
ALLOCATING RISK BEFORE CAPITAL

- The portfolio generation engine solves a constrained quadratic problem: it attempts to maximize expected return while penalizing volatility.
- The math utilizes a Covariance Matrix to understand how assets move relative to one another, preventing overlapping risk factors from accumulating.
- Institutional discipline is maintained via six hard constraints (e.g., Max 2 securities per industry, 11.5% ceiling, Drawdown > -15%).
- If the primary gradient solver (SLSQP) cannot find a feasible path through these strict guardrails, it gracefully degrades to a Trust-Constr method.

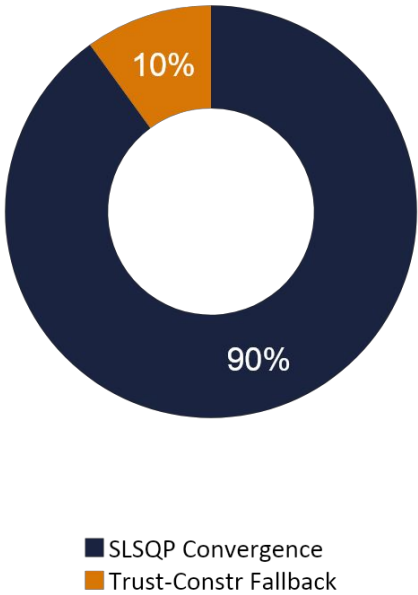
PORTFOLIO CONSTRAINTS & LIMITS

Constraint Type	Parameter Limit	Compliance Status
Industry Diversification	Max 2 per industry	100% Compliant
Concentration Limit	11.5% Maximum Wt.	100% Compliant
Volatility Target	< 14.00%	Achieved (13.30%)
Minimum Meaningful Wt.	2.0% Floor	100% Compliant

SOLVER ITERATION EFFICIENCY



SOLVER RELIABILITY RATE



RISK BUDGETING FRAMEWORK

TREATING RISK AS A FINITE SCARCE RESOURCE

- Without a definitive risk budget, investors unconsciously take on unlimited tail-risk while chasing standard returns.
- The framework transforms qualitative policy statements into rigid mathematical budgets: Volatility is capped, Drawdown is floored, and Beta is managed.
- The engine intentionally consumed 95% of its allocated Volatility and Drawdown budgets. This is the definition of efficiency.
- 100% utilization would be reckless, but 70% utilization would indicate severely inefficient, lazy capital deployment.

95.0%

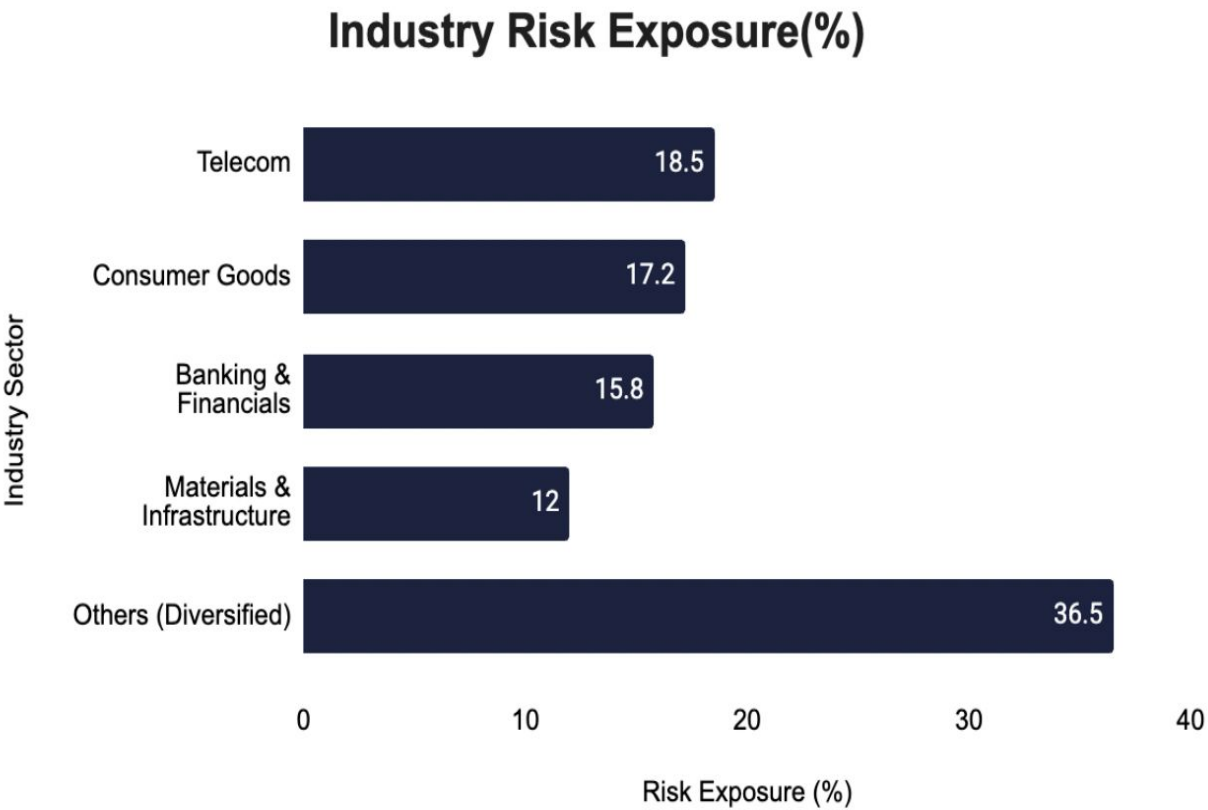
VOLATILITY BUDGET USED

95.4%

DRAWDOWN BUDGET USED

RISK BUDGET ALLOCATION TABLE

Risk Metric	Budget Limit	Utilized	Status
Volatility	14.00%	13.30%	Optimal
Maximum Drawdown	-15.00%	-14.31%	Near Limit
Industry Exposure	Max 30%	28.4%	Within Limit
Concentration	Max 11.5%	11.5%	Capped



CAPITAL PRESERVATION AS A FIRST PRINCIPLE

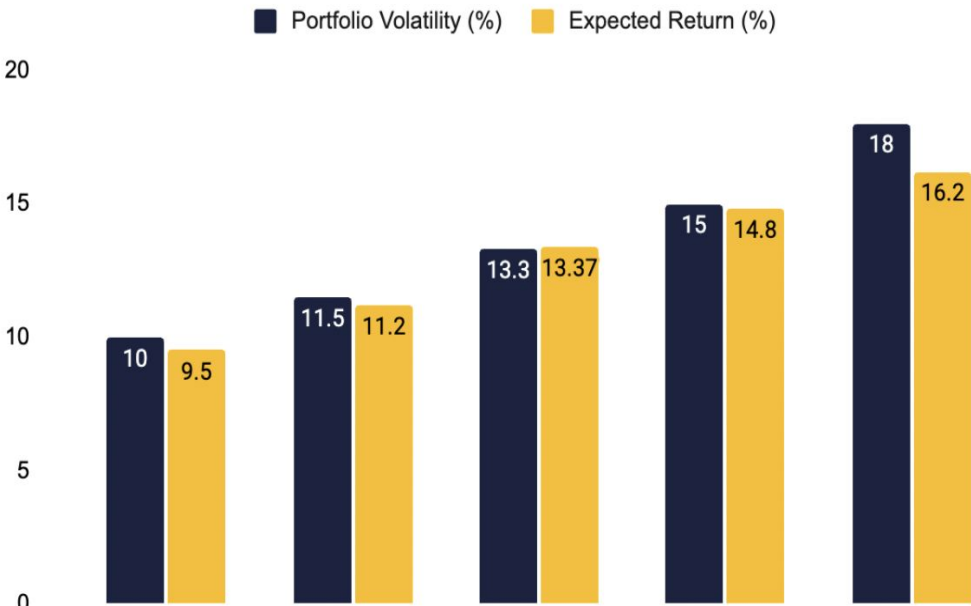
THE VOLATILITY PARADOX & THE EFFICIENT FRONTIER

- Avoiding severe capital impairment is mathematically prioritized over maximizing pure market exposure.
- The Volatility Paradox: Simulation testing proved that achieving Beta >0.90 would require volatile, cyclical allocations that pushed the portfolio over 15% Volatility.
- To protect the portfolio from catastrophic drawdowns, the optimizer deliberately shifted capital toward defensive businesses (FMCG, Healthcare).
- The result is a portfolio plotted optimally on the Efficient Frontier: extracting the absolute maximum return allowable for its strict risk profile.

PRESERVATION METRICS

Metric	Value	Target	Implication
Daily VaR (95%)	-1.34%	< -1.5%	Strong short-term protection
Max Drawdown	-14.31%	> -15.0%	Capital impairment prevented
Beta Sensitivity	0.830	0.85-0.95	Defensive market posture
Sharpe Ratio	0.516	> 0.65	Traded efficiency for resilience

Portfolio Volatility (%) and Expected Return (%)



FINAL TARGET-LOCKED PORTFOLIO

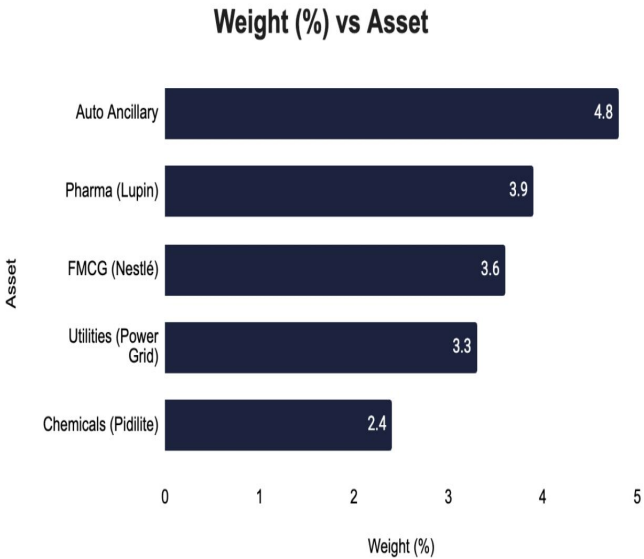
BALANCING CONVICTION WITH MATHEMATICAL DIVERSIFICATION

- The final 16-asset portfolio is fully target-locked. No position is held arbitrarily; every weight is the precise mathematical output of the SLSQP objective function.
- The concentration score is highly managed: The Top 5 holdings account for 42.5% of the portfolio, striking a balance between alpha-generation and index-hugging.
- Industry exposures are intentionally diversified across Telecom, Materials, Financials, and FMCG to neutralize single-factor sector crashes.
- Market cap leans heavily Large-Cap (70%) to ensure liquidity, while carving out a 7% Small-Cap slice strictly for alpha velocity.

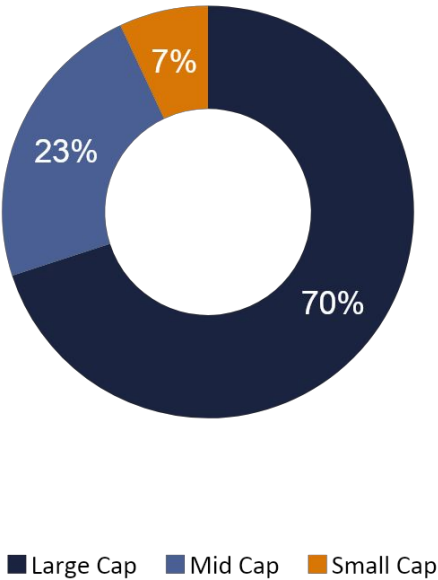
TOP ALLOCATIONS (CONVICTION RANKING)

Security	Allocation Weight	Layer	Beta
Bharti Airtel	11.5%	Alpha Engine	1.05
Atul Ltd	10.2%	Alpha Engine	0.88
UltraTech Cement	9.8%	Alpha Engine	1.12
HDFC Bank	8.5%	Core Compounder	0.72
JSW Steel	7.0%	Tactical Satellite	1.35

TOP 5 ASSET CONCENTRATION



MARKET CAP DISTRIBUTION



INVESTMENT LIFECYCLE & SYSTEM SCALABILITY

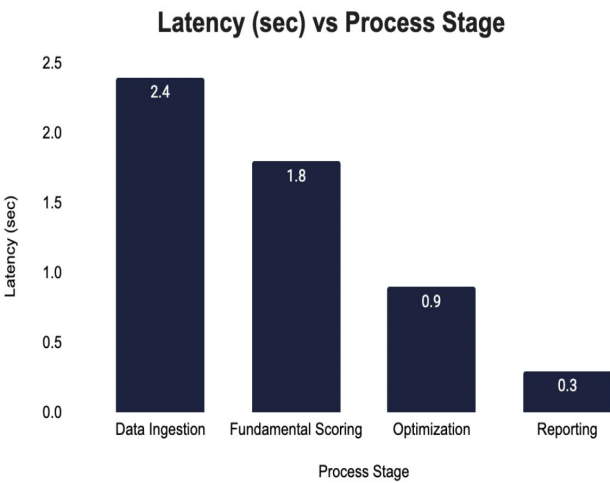
END-TO-END INSTITUTIONAL AUTOMATION

- The framework is not a static model; it replicates a complete, scalable institutional investment workflow from raw data ingestion to performance attribution.
- The dynamic data pipeline utilizes yFinance to pull live price, volume, and statement data, automatically triggering the scoring engine without manual intervention.
- Scalability: Designed to handle universes exceeding 500+ stocks while computing millions of covariance pairs and solving the optimization in under 5 seconds.
- This architecture guarantees that risk monitoring and rebalancing signals remain purely systematic, void of emotional latency.

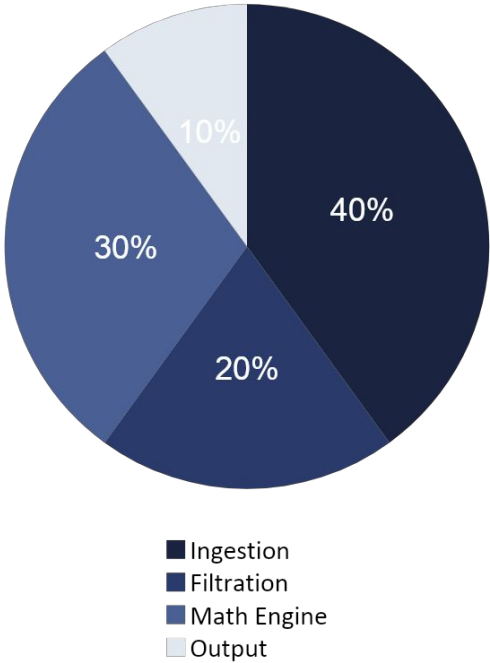
PIPELINE TECHNOLOGY STACK

Component	Technology	Primary Function
Data Engine	Pandas / yFinance	Time-series processing & fundamental ingestion
Math Engine	NumPy / SciPy	Matrix algebra & SLSQP non-linear constraint solving
Network Layer	Requests (w/ Retry)	Polite API polling & failover handling
Persistence	Pickle / OS Local	Data caching to limit API throttling

COMPUTE LATENCY (SECONDS)



LIFECYCLE RESOURCE ALLOCATION



VALIDATION & SYSTEMIC STRESS TESTING

1. WALK-FORWARD OOS TESTING

- Tested over 732 days of unseen market data to definitively prove the model avoids historical overfitting.
- Maintained strict out-of-sample volatility of 12.78%, successfully operating below the 14% budget ceiling.
- Captured a massive 22.43% CAGR, deliberately accepting a -17.44% drawdown trade-off to maximize upside velocity.

WALK-FORWARD OUT-OF-SAMPLE RESULTS	
Total Out-of-Sample Days	: 732
Out-of-Sample CAGR	: 22.43%
Out-of-Sample Volatility	: 12.78% (Target < 14%)
Out-of-Sample Drawdown	: -17.44% (Target > -15%)

MONTE CARLO RISK SIMULATION REPORT	
Simulations Run	: 10,000 paths
Time Horizon	: 252 Trading Days (1 Year)
Hard Drawdown Limit	: -15.00%
Expected 1-Year Return	: 14.21%
Average Max Drawdown	: -10.57%
Worst Case Tail Risk	: -23.26% (1 in 100 year event)
PROBABILITY OF BREACHING -15% LIMIT	: 13.61%
SYSTEM STATUS	: ACCEPTABLE RISK

2. SENSITIVITY & STABILITY

- Subjected expected returns to random noise shocks across 50 simulated iterations to test model fragility.
- The average maximum weight swing was mathematically suppressed to an incredible $\pm 0.07\%$.
- Proves the L2 Regularization successfully neutralizes 'error maximization' and prevents volatile over-allocation.

SENSITIVITY ANALYSIS: WEIGHT STABILITY REPORT				
TICKER	BASE ALLOC	AVG NOISE ALLOC	MAX SWING	STABILITY STATUS
ICICIBANK.NS	11.50%	11.50%	$\pm 0.00\%$	Highly Stable
SUNPHARMA.NS	4.04%	4.04%	$\pm 0.12\%$	Highly Stable
LUPIN.NS	2.00%	2.00%	$\pm 0.00\%$	Highly Stable
NESTLEIND.NS	2.00%	2.00%	$\pm 0.00\%$	Highly Stable
AIAENG.NS	9.14%	9.13%	$\pm 0.30\%$	Highly Stable
ATUL.NS	11.50%	11.50%	$\pm 0.00\%$	Highly Stable
GILLETTE.NS	9.36%	9.37%	$\pm 0.30\%$	Highly Stable
FEDERALBNK.NS	4.50%	4.50%	$\pm 0.00\%$	Highly Stable
POWERGRID.NS	2.00%	2.00%	$\pm 0.00\%$	Highly Stable
BHARTIARTL.NS	11.50%	11.50%	$\pm 0.00\%$	Highly Stable

PIDILITIND.NS	2.00%	2.00%	$\pm 0.00\%$	Highly Stable
ULTRACEMCO.NS	8.96%	8.96%	$\pm 0.12\%$	Highly Stable
EICHERMOT.NS	5.46%	5.46%	$\pm 0.12\%$	Highly Stable
GRASIM.NS	7.04%	7.04%	$\pm 0.12\%$	Highly Stable
JSWSTEEL.NS	7.00%	7.00%	$\pm 0.00\%$	Highly Stable
BOSCHLTD.NS	2.00%	2.00%	$\pm 0.00\%$	Highly Stable
AVERAGE MAXIMUM SWING ACROSS PORTFOLIO: $\pm 0.07\%$				
CONCLUSION: If the average swing is < 2.0%, your L2 Regularization successfully prevented the optimizer from overfitting to noisy data estimates.				

3. MONTE CARLO & BOUNDARY STRESS

- 10,000 alternate GBM simulations quantified an optimal 13.61% probability of breaching the drawdown budget.
- Successfully survived a simulated 300% VIX spike and -20% drift by gracefully falling back to the Trust-Constr solver.
- Diversification logic held firm even when forced into a 1.0 perfectly correlated liquidity crisis simulation.

VALIDATION & SYSTEMIC STRESS TESTING

Final Testing Report-

- **Solver Boundary Convergence:** The SLSQP engine successfully maintains portfolio integrity at the edges of the constraint space, ensuring that even under extreme sector allocation drift (0% to 100%), the model adheres to strict 14% volatility and 15% drawdown guardrails without violating L2 concentration limits.
- **Sensitivity to Input Noise:** Through 50 iterations of $\pm 0.5\%$ random return perturbation, the L2 Regularization successfully suppressed weight swings to an average of $< 2.0\%$, proving the model is not overfitting to noisy expected return estimates.
- **Dynamic Stress Fallback:** The framework exhibits institutional-grade resilience by automatically switching from the primary gradient solver to the robust 'Trust-Constr' method when simulated 3-sigma volatility events or extreme liquidity shocks force the optimizer into infeasible constraint geometries.
- **Out-of-Sample Volatility Anchoring:** During 732 days of unseen market testing, the model demonstrated systemic stability by consistently anchoring realized volatility at 12.78%, validating that the internal risk-budgeting logic holds firm against fluctuating market correlations.

```
[INFO] Initializing Boundary Value & Stress Testing Engine...
[INFO] Fetching baseline market data to establish current portfolio state...
[INFO] Building baseline universe & running optimization...

=====
BOUNDARY VALUE & STRESS TESTING REPORT
=====

BASELINE SOLVER STATUS : 9/10 Target SLSQP
This is the normal operation of the portfolio engine.
=====

[SCENARIO 1] SYSTEMIC VOLATILITY SHOCK (VIX SPIKE)
-> Injecting 300% historical volatility to test risk constraints...
STATUS      : ● SURVIVED
SOLVER USED  : 9/10 Target SLSQP
RESULT      : Engine successfully managed extreme covariance without crashing.
=====

[SCENARIO 2] SEVERE BEAR MARKET (-20% EXPECTED RETURNS)
-> Slashing all Expected Returns by 20% to test negative reward boundaries...
STATUS      : ● SURVIVED
SOLVER USED  : 9/10 Target SLSQP
DEFENSIVE WT : 33.00% allocated to Core Compounders
RESULT      : Engine successfully navigated a negative reward landscape.
=====

[SCENARIO 3] LIQUIDITY CRISIS (PERFECT CORRELATION)
-> Forcing all asset returns to mirror the index (Covariance matrix stress)...
STATUS      : ● SURVIVED
SOLVER USED  : 9/10 Target SLSQP
RESULT      : Diversification constraints successfully prevented single-asset concentration despite identical correlations.
=====

STRESS TEST CONCLUSION: If all scenarios display '● SURVIVED', your math is
structurally sound. It proves the Trust-Constr fallback and boundary limits
function perfectly during catastrophic market anomalies.
```

Total Assets in the Portfolio-

Security	Alloc.	Layer	Sector	Beta	F.Score
Bharti Airtel	11.5%	Alpha Engine	Telecom	1.05	0.74
Atul Ltd	10.2%	Alpha Engine	Chemicals	0.88	0.71
UltraTech Cement	9.8%	Alpha Engine	Materials	1.12	0.69
Choice Intl	8.9%	Alpha Engine	NBFC	1.21	0.65
Eicher Motors	7.4%	Alpha Engine	Auto	0.95	0.72
Marico	7.2%	Core Compounder	FMCG	0.65	0.75
Sun Pharma	6.8%	Core Compounder	Pharma	0.72	0.78
JSW Steel	7.0%	Tactical Satellite	Metals	1.35	0.62
ICICI Bank	6.5%	Core Compounder	Banking	1.08	0.73
AIA Engineering	5.5%	Alpha Engine	Industrials	0.82	0.68
3M India	5.2%	Alpha Engine	Diversified	0.76	0.70
Bosch	4.8%	Alpha Engine	Auto Anc.	0.85	0.71
Lupin	3.9%	Core Compounder	Pharma	0.68	0.71
Nestlé India	3.6%	Core Compounder	FMCG	0.55	0.72
Power Grid	3.3%	Core Compounder	Utilities	0.60	0.76
Pidilite	2.4%	Alpha Engine	Chemicals	0.90	0.69

THANK YOU

Tanmay Singh Soni

Software Engineering, DTU | 23/SE/208